

Built-in R Features - Math

We've talked a bit about some of the built-in math functions and features in R, but let's have one more look at a few of them:

- `abs()`: computes the absolute value.
- `sum()`: returns the sum of all the values present in the input.
- `mean()`: computes the arithmetic mean.
- `round()`: rounds values (additional arguments to nearest)

Here's a quick example of each:

In [7]:

```
v <- c(-1,0,1,2,3,4,5)
```

In [8]:

```
abs(-2)
```

Out[8]:

2

In [9]:

```
abs(v)
```

Out[9]:

1 0 1 2 3 4 5

In [10]:

```
sum(v)
```

Out[10]:

14

In [11]:

```
mean(v)
```

Out[11]:

2

In [13]:

```
round(23.1231)
```

Out[13]:

23

In [14]:

```
round(23.1231234,2)
```

```
Out[14]:
```

```
23.12
```

Want more built-in math functions? Check out:

Numeric Functions

Function	Description
abs(<i>x</i>)	absolute value
sqrt(<i>x</i>)	square root
ceiling(<i>x</i>)	ceiling(3.475) is 4
floor(<i>x</i>)	floor(3.475) is 3
trunc(<i>x</i>)	trunc(5.99) is 5
round(<i>x</i>, digits=<i>n</i>)	round(3.475, digits=2) is 3.48
signif(<i>x</i>, digits=<i>n</i>)	signif(3.475, digits=2) is 3.5
cos(<i>x</i>), sin(<i>x</i>), tan(<i>x</i>)	also acos(<i>x</i>), cosh(<i>x</i>), acosh(<i>x</i>), etc.
log(<i>x</i>)	natural logarithm
log10(<i>x</i>)	common logarithm
exp(<i>x</i>)	e^x

You can also check out the documentation for even more functions, like those related to statistics (we'll cover those later on).